

```
'''
Numpy : is a general purpose array processing library for python
it provides high performance multidimensional array object along with a
collection
of tools for working efficiently with arrays

operations on numpy
1. perform operations in one and multidimensional arrays
2. perform operations of vector
3. broadcasting capabilities for working with arrays in different shapes
4. mathematical ,statistical and linear algebra functions
what do you mean by general purpose
library or tool or soft to be useful for a wide variety of tasks
than being limited to one specific purpose....
'''
```

```
import numpy as np
```

```
# what is arrays : arrays is a collection of similar type of elements or it is collecti
```

```
# create arrays
arr = np.array([1,2,3])
```

```
print(arr)
```

```
[1 2 3]
```

```
# shape of arrays : number of dimensions rank size of dimension shape
```

```
arr = np.array([1,2,3,4,5])
print("original array : ",arr)
print("shape of array : ",arr.shape)
```

```
print("data type of array : ",arr.dtype)
print("number of dim of array : ",arr.ndim)
```

```
original array : [1 2 3 4 5]
shape of array : (5,)
data type of array : int64
number of dim of array : 1
```

## # Creating 2D Arrays

```
arr = np.array([[1,2,3],[2,3,4],[3,6,8]])
print("original array : ",arr)
print("shape (order of mat ) of array : ",arr.shape)
print("data type of array : ",arr.dtype)
print("number of dim of array : ",arr.ndim)
```

```
original array : [[1 2 3]
 [2 3 4]
 [3 6 8]]
shape (order of mat ) of array : (3, 3)
data type of array : int64
number of dim of array : 2
```

## # without values , special arrays

```
arr = np.zeros((3,4))
print(arr)
print(arr.size)
print(arr.shape)
```

```
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
12
(3, 4)
```

```
arr = np.ones((3,4))
print(arr)
print(arr.size)
print(arr.shape)
```

```
[[1. 1. 1. 1.]
 [1. 1. 1. 1.]
 [1. 1. 1. 1.]]
12
(3, 4)
```

```
# identity matrix
arr = np.eye(3)
print(arr)
```

```
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

```
# range
arr = np.arange(0,20,2)
print(arr)
```

```
[ 0  2  4  6  8 10 12 14 16 18]
```

```
# linspace
arr = np.linspace(1,100,5)
print(arr)
```

```
[ 1.    25.75  50.5   75.25 100. ]
```

```
# indexing and slicing
arr = np.array([10,20,30,40,50])
try:
    print(arr[0])
    print(arr[2])
    print(arr[5])
except IndexError as e:
    print(e)
```

```
10
30
index 5 is out of bounds for axis 0 with size 5
```

```
arr = np.array([10,20,30,40,50])
print(arr[1:4])
```

```
[20 30 40]
```

```
# two dimensional array  
arr = np.array([  
    [1,2,3],  
    [4,5,6],  
    [7,8,9]  
])
```

```
# access the elements using index  
print(arr[2][2])
```

9

```
# entire columns  
print(arr[0:2,1])
```

[2 5]

```
# entire rows  
print(arr[1,:])
```

[4 5 6]